**Exploring Fundamentals of Microstructural Design**

An exploration of:

Conversion Coatings on Non-Ferrous Metals

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# Introduction

## Corrosion Issues for Non-ferrous Metals

Non-ferrous metals such as aluminum and its alloys, magnesium, zinc, and their coated steels are widely used for transportation, aerospace, construction, and consumer goods because of their high strength-to-weight ratio, formability, and conductivity. Despite forming native oxide films, these metals can suffer significant corrosion in humid, marine, or chemically aggressive environments, leading to loss of mechanical integrity, aesthetic degradation, and premature failure. To meet stringent durability and safety requirements, industries typically rely on multilayer coating systems in which an inorganic conversion layer is a critical pretreatment step beneath paints, sealants, or adhesives [1].

## What is a Conversion Coating?

In surface treatment, a conversion coating is a thin layer formed on a metal surface through a chemical or electrochemical reaction that converts the outermost metal or its oxide into an adherent, insoluble compound. During this conversion process, metal ions from the substrate and anions from the treatment bath react to produce a new compound layer that is strongly bonded to the base metal, improving corrosion resistance and providing a better key for subsequent organic coatings. Conversion coatings may be generated purely chemically (e.g. chromate or phosphate) or electrochemically (e.g. anodizing) and are essential to acheive robust paint adhesion and long-term protection in demanding applications [1].

Five major families dominate conversion coating technology for non-ferrous and coated steels: chromate conversion coatings, anodizing of aluminum, zirconium/titanium-based nano-ceramic coatings, silane-based treatments, and phosphate conversion coatings. Each family differs in deposition mechanism, film structure and thickness, typical process steps, and performance trade-offs in terms of corrosion resistance, adhesion promotion, environmental profile, and cost. The following sections examine these systems with emphasis on processing parameters, application areas, and the trends driving a shift toward chromium-free, nano-scale, and hybrid pretreatment stacks.

# Chromate Conversion

## What is it?

Chromate conversion coatings (CCCs) have historically been the benchmark pretreatment for aluminum and other non-ferrous substrates in aerospace and defense because they combine very good corrosion resistance with a unique self-healing capability. These coatings incorporate hexavalent chromium species, such as chromium and dichromate, which remain partly soluble within the coating and can migrate into damaged areas, where they are reduced and precipitate as protective chromium compounds that repassivate exposed

metal. As a result, CCCs can inhibit both anodic and cathodic reactions, providing robust protection even when the coaring is scratched or partially damaged.

### The Process

CCCs are typically applied to cleaned aluminum or other non-ferrous substrates by immersion or spray in acidic chromate-bearing solutions, followed by rinsing and drying. A conventional process sequence consists of alkaline cleaning or degreasing to remove organic soils, a deoxidizing or desmut step if needed, immersion or spray in a chromate conversion bath for a controlled time, water rinsing, and oven or air drying to fix the film. Key parameters include bath pH (often mildly acidic), temperature, time, and chromate concentration, which together determine coating thickness, color, and corrosion performance. Proper control is required to obtain uniform, adherent film without excessive powdery buildup.

### Use Cases

The outstanding performance of hexavalent chromate systems is overshadowed by the high toxicity and carcinogenicity of CR(VI) compounds, which has driven strong regulatory pressure under frameworks such as REACH and similar national regulations. Many countries have restricted or phased out traditional CR(VI)-based treatments in favor of trivalent chromium systems or completely chromuim-free alternatives for most applications, although exemptions persist in some aerospace and defense uses where equivalent performance has been difficult to match. Trivalent chromate conversion coatings and nanoparticle-modified trivalent systems have been developed that can achieve good corrosion resistance and even some degree of self-repair, but they generally require careful process optimization and may not fully replicate the robustness of legacy hexavalent coatings in every environment.

## Anodizing of Aluminum

### What is it?

Anodizing is an electrochemical conversion process in which the aluminum component is made the anode in an electrolytic cell so that its surface is oxidized to a thick, adherent layer of porous aluminum oxide. Unlike chemical conversion coatings that rely purely on chemical reactions, anodizing uses applied current or voltage to control oxide growth in electrolytes such as sulfuric, chromic, oxalic, or phosphoric acid. The resulting anodic film is typically non-conductive, has well-defined porosity and thickness, and provides an excellent base for coloring and sealing treatments that significantly enhance corrosion and wear resistance.

## The Process

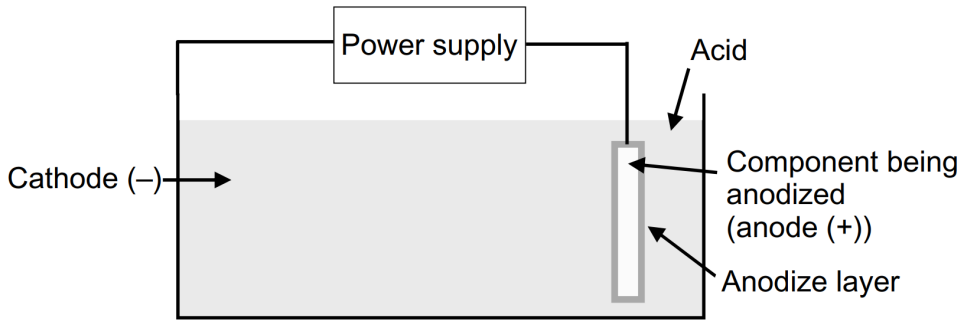


**Figure X: Schematic of the Anodizing Process** [1]

A typical anodizing line for aluminum begins with cleaning and degreasing to remove oils, followed by alkaline etching to adjust surface roughness and appearance, and a desmut step to remove intermetallic residues. The parts are then anodized in an acid electrolyte under controlled voltage or current density and temperature, with process recipes varying for sulfuric, chromic, and oxalic acid systems. After anodizing, the porous oxide layer may be dyed or impregnated with pigments for decorative finishes and is usually sealed in hot water or nickel-containing solutions to close the pores and improve corrosion resistance.

Operating parameters such as acid concentration, bath temperature, current density, and time strongly influence oxide thickness, pore morphology, hardness, and color. Sulfuric acid anodizing (often termed Type II) is the most widely used process for general corrosion protection and decorative coloring, while low-temperature, high-current-density sulfuric processes produce thicl "hardcoat" anodic films with high wear resistance (Type III) [2]. Chromic acid anodizing has been important in aerospace because of its relatively thin but highly protective films, although environmental concerns over chromic acid have also prompted moves toward sulfuric and mixed-acid alternatives.

## Use Cases

Anodizing is extensively used for decorative and architectural aluminum, where colored and sealed oxide layers provide both appearance and durability for façade elements, consumer products, and electronics housings. Hard anodizing is applied to components requiring high abrasion resistance, such as hydraulic cylinders, piston surfaces, and machine parts. In aerospace, anodizing often competes with chromate conversion coatings, with chromic or sulfuric anodizing used on structural aluminum parts as a base for primers and paints, while CCCs are favored where electrical conductivity or minimal dimensional change is needed. In many cases, anodizing and conversion coatings are complimentary options within the same platform, selected based on geometry, required conductivity, fatigue performance, and coating system design [2].

# Zirconium/Titanium "Nano-ceramic" Conversion

### What is it?

### The Process

### Use Cases

# Silane-based Coatings

### What is it?

### The Process

### Use Cases

# Phosphate Conversion Coatings

### What is it?

### The Process

### Use Cases

# Comparison, Application, and Trends

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